

Universal Windows Platform

UWP is a layer of APIs that are guaranteed by Microsoft to be available on any device that is running Windows, whether it be a mobile phone, a PC, the Xbox, an embedded device like the Raspberry Pi or even a device that hasn't been released yet.

There are still platform-specific features, that can be layered on top of the UWP via extension SDKs. These SDKs extend the feature set of Windows for a particular family of devices. For example, the mobile family of Windows might power phones and phablets, while the desktop family on Windows will power tablets, laptops and desktops. They both have their own extension SDKs that can be used to implement platform-specific features. Microsoft estimates though that over 85% of Windows APIs are available universally.

The same application can now support multiple device families more easily, not just by using only the common UWP APIs, but by detecting and using only the features available on that platform. There is now an API to check for the presence / absence of platform-specific features and APIs.

This is different from earlier versions when such checks could only be performed at compile time in order to build different binaries for different platforms. Now runtime checks allow developers to have the same application code adapt to mobile or desktop.

Consider that Windows 10 now makes it possible to have a phone power a full desktop experience if docked to a station that has a larger screen, a keyboard and mouse. Of course this will need a very powerful phone, and might not be feasible today, but the possibility of such a thing is eminent. Microsoft's new modular approach means that the OS for such a phone could simply be a composition of the Universal Windows Platform with Mobile and Desktop extensions.

With such future devices it will be important for applications to adapt at runtime from a mobile form factor, to a desktop form factor, to even a television form factor. For this Microsoft has provided new adaptive UI elements that can work across form factors. Developers can also embrace the principles of responsive design for desktop applications – already a popular approach for web apps.

Universal Windows Platform Bridges

It's quite a mouthful, but these bridges – as we will refer to them as for brevity – are toolkits designed to make it easier for developers to bring